

Triangles, radians and sectors - Set 2 - Solutions

Specification MM4.1, MM4.2, M5.16.

Answer key

Question	Answer	Topic
1	A	Triangles, radians and sectors
2	E	Triangles, radians and sectors
3	D	Triangles, radians and sectors
4	D	Triangles, radians and sectors
5	C	Triangles, radians and sectors

Worked solutions

Question 1 TRI-11 · Triangles, radians and sectors · spec MM4.2

A circle has radius r . A sector of angle θ radians has the same area as the segment cut off by a chord that subtends an angle of $\frac{\pi}{2}$ at the centre. Find θ .

Answer: A

$$\text{Segment area} = \frac{1}{2}r^2\left(\frac{\pi}{2} - \sin \frac{\pi}{2}\right) = \frac{1}{2}r^2\left(\frac{\pi}{2} - 1\right). \text{ Sector area } \frac{1}{2}r^2\theta. \text{ So } \theta = \frac{\pi}{2} - 1.$$

Question 2 TRI-12 · Triangles, radians and sectors · spec MM4.1

A triangle has sides a, b, c with $a^2 + b^2 < c^2$. Which of the following statements is/are true?

- I The angle opposite c is obtuse.
- II c is the longest side.
- III The triangle cannot be isosceles.

Answer: E

Cosine rule: $\cos C = \frac{a^2 + b^2 - c^2}{2ab} < 0$, so C is obtuse (I). The largest angle is opposite the longest side, so c is longest (II). III is false: $a = b = 1, c = 1.5$ works.

Question 3 TRI-14 · Triangles, radians and sectors · spec MM4.1

A vertical pole BT stands on horizontal ground. From a point A on the ground the angle of elevation of T is 30° ; from a point C on the ground, 20 m closer to the pole than A and in line with A and B , the angle of elevation is 60° . Find the height of the pole.

Answer: D

$$\text{Let the height be } h. AB = h\sqrt{3} \text{ and } CB = \frac{h}{\sqrt{3}}. AB - CB = 20 \Rightarrow h\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \frac{2h}{\sqrt{3}} = 20 \Rightarrow h = 10\sqrt{3}.$$

Question 4 TRI-15 · Triangles, radians and sectors · spec MM4.1

Triangle ABC has $AB = 7, BC = 8, CA = 9$. Find the area of the triangle.

Answer: D

$$\cos A = \frac{49 + 81 - 64}{2 \cdot 7 \cdot 9} = \frac{66}{126} = \frac{11}{21}, \text{ so } \sin A = \frac{\sqrt{441 - 121}}{21} = \frac{\sqrt{320}}{21} = \frac{8\sqrt{5}}{21}. \text{ Area} = \frac{1}{2} \cdot 7 \cdot 9 \cdot \frac{8\sqrt{5}}{21} = 12\sqrt{5}.$$

Question 5 TRI-16 · Triangles, radians and sectors · spec MM4.2

Two circles of radius 2 have their centres a distance 2 apart. Find the area of the region common to both circles.

Answer: C

The common chord subtends $\frac{2\pi}{3}$ at each centre (equilateral triangles of side 2). The overlap is two segments:
 $2 \times \frac{1}{2} \cdot 4\left(\frac{2\pi}{3} - \sin \frac{2\pi}{3}\right) = 4\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right) = \frac{8\pi}{3} - 2\sqrt{3}.$